

Team_5071_1

Database Design Assignment

Parts 2 & 3: Dataset Selection and ER Diagram

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PART 2: Dataset Selection

Dataset Information

Dataset Name: Historical Global Seismic Events Database (1826-2026)

Source: Kaggle

Format: CSV (Comma-Separated Values)

Size: 106,077 records × 19 attributes

Coverage: Global seismic events from 1826 to January 2026

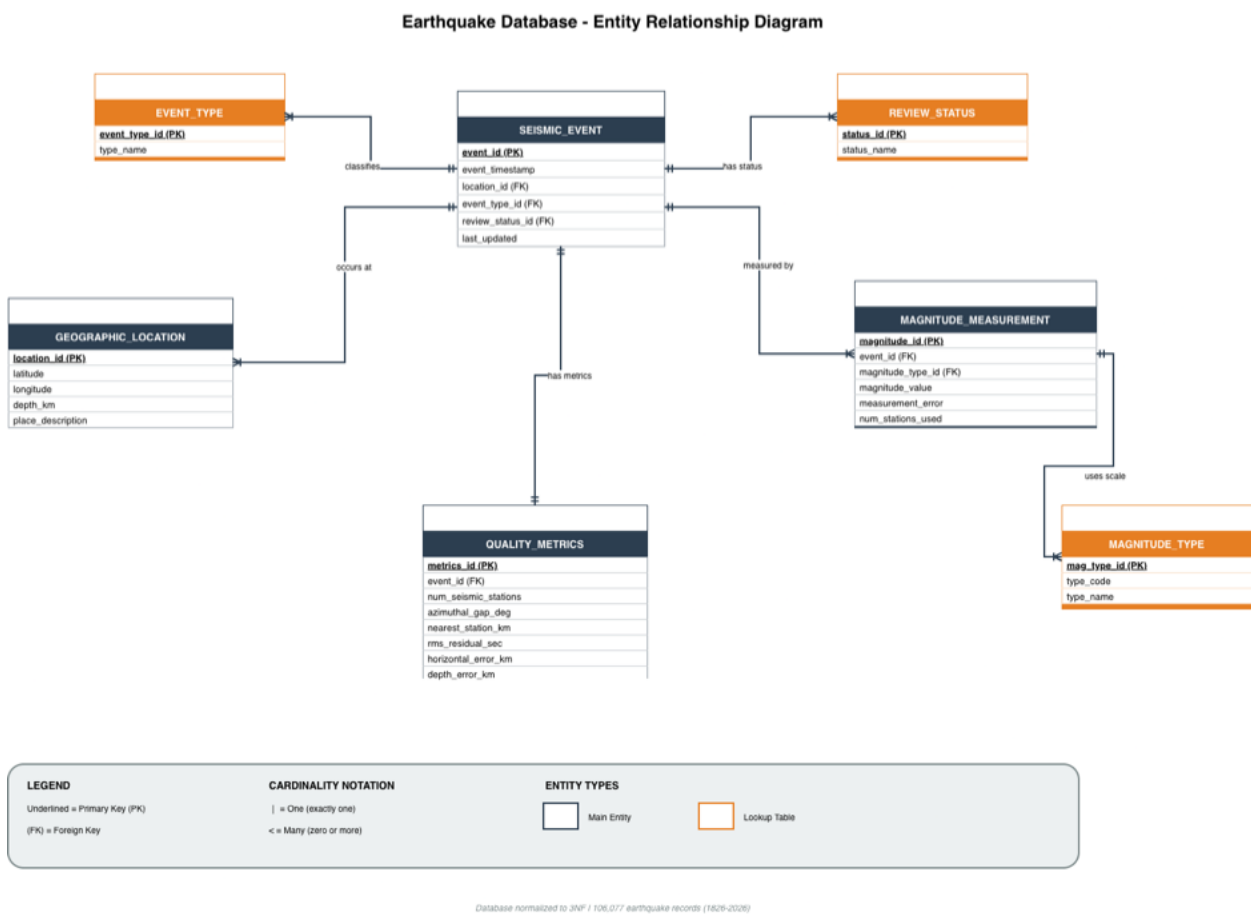
URL: <https://www.kaggle.com/dhrubangtalukdar/200-years-of-global-major-earthquakes-18262026?resource=download>

Justification

We selected the earthquake dataset from Kaggle because it provides database design experience with real-world data. The dataset covers nearly 200 years of global seismic events with over 106,000 records, offering substantial data volume for SQL queries. The data naturally decomposes into multiple entities including seismic events, geographic locations, magnitude measurements, and quality metrics, with well-defined relationships, making it a good candidate for demonstrating third normal form (3NF) database design principles. Additionally, the presence of 28 magnitude types, 7 event types, and repeating geographic locations provides opportunities to showcase proper use of foreign keys, lookup tables, and relationship cardinalities. This dataset is comprehensive enough that one could use a data science approach to analyzing natural disasters.

PART 3: Entity-Relationship Diagram Design

ERD Image



Database Schema Overview

Our normalized database design consists of 7 entities organized into two categories:

- 4 Main Entities: SEISMIC_EVENT, GEOGRAPHIC_LOCATION, MAGNITUDE_MEASUREMENT, QUALITY_METRICS
- 3 Lookup Tables: EVENT_TYPE, REVIEW_STATUS, MAGNITUDE_TYPE

Entity Definitions

1. SEISMIC_EVENT

Purpose: Core table representing individual seismic occurrences

Attribute	Key	Description
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event_id	PK	Unique USGS identifier for each event
event_timestamp		Date and time of occurrence (UTC)
location_id	FK → GEOGRAPHIC_LOCATION	Reference to location
event_type_id	FK → EVENT_TYPE	Reference to event classification
review_status_id	FK → REVIEW_STATUS	Data quality status
last_updated		Last modification timestamp

Rationale: Central entity from which all other entities are referenced. Foreign keys enforce referential integrity and enable joins.

2. GEOGRAPHIC_LOCATION

Purpose: Geographic coordinates and location descriptions

Attribute	Key	Description
location_id	PK	Unique location identifier
latitude		Latitude in decimal degrees
longitude		Longitude in decimal degrees
depth_km		Depth below surface in kilometers
place_description		Human-readable location name

Rationale: Separated to reduce redundancy. Multiple earthquakes can occur at similar coordinates.

3. MAGNITUDE_MEASUREMENT

Purpose: Magnitude readings and measurement metadata

Attribute	Key	Description
magnitude_id	PK	Unique measurement identifier
event_id	FK → SEISMIC_EVENT	Event being measured
magnitude_type_id	FK → MAGNITUDE_TYPE	Magnitude scale used
magnitude_value		Numeric magnitude value
measurement_error		Uncertainty in magnitude
num_stations_used		Stations contributing to calculation

Rationale: Supports multiple magnitude measurements per event (different agencies/scales).

4. QUALITY_METRICS

Purpose: Measurement quality and network coverage indicators

Attribute	Key	Description
metrics_id	PK	Unique metrics identifier
event_id	FK → SEISMIC_EVENT	Associated event
num_seismic_stations		Total reporting stations
azimuthal_gap_deg		Largest gap in station coverage
nearest_station_km		Distance to nearest station
rms_residual_sec		RMS of time residuals
horizontal_error_km		Horizontal location uncertainty
depth_error_km		Depth uncertainty

Rationale: Quality metrics are functionally dependent on events but conceptually distinct. Separation improves clarity and handles NULL values (one can see in the data that past historical records lack these measurements).

5. EVENT_TYPE (Lookup Table)

Purpose: Classification of seismic event types

Attribute	Key	Description
event_type_id	PK	Unique type identifier
type_name		Event type name
type_description		Detailed description

Valid Values: earthquake, landslide, volcanic eruption, nuclear explosion, mine collapse, rock burst, explosion

Rationale: Normalization to 3NF. Only 7 distinct values in dataset. Eliminates string repetition and enables adding type metadata.

6. REVIEW_STATUS (Lookup Table)

Purpose: Data quality review status

Attribute	Key	Description
status_id	PK	Unique status identifier
status_name		Status name
status_description		Status meaning

Valid Values: reviewed, automatic

Rationale: Normalization to 3NF. Enables adding status workflow information.

7. MAGNITUDE_TYPE (Lookup Table)

Purpose: Magnitude scale definitions

Attribute	Key	Description
mag_type_id	PK	Unique scale identifier
type_code		Short code (mb, mw, etc.)
type_name		Full scale name

type_description		Calculation methodology
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Common Values: mb (body wave), mw (moment), mww (W-phase moment), mwc (centroid moment), ms (surface wave), etc. (28 total)

Rationale: Normalization to 3NF. 28 distinct magnitude types. Reduces redundancy and enables adding scientific metadata about each scale.

Relationship Mappings

Relationship 1: SEISMIC_EVENT → EVENT_TYPE

- **Cardinality:** Many-to-One (M:1)
- **Description:** Each seismic event has exactly one classification type. Each type can classify many events.
- **Implementation:** event_type_id foreign key in SEISMIC_EVENT
- **Business Rule:** All events must be classified (NOT NULL constraint)

Relationship 2: SEISMIC_EVENT → REVIEW_STATUS

- **Cardinality:** Many-to-One (M:1)
- **Description:** Each event has exactly one review status. Each status applies to many events.
- **Implementation:** review_status_id foreign key in SEISMIC_EVENT
- **Business Rule:** All events have a status (NOT NULL constraint)

Relationship 3: SEISMIC_EVENT → GEOGRAPHIC_LOCATION

- **Cardinality:** Many-to-One (M:1)
- **Description:** Each event occurs at one location. Each location may have multiple events over time.
- **Implementation:** location_id foreign key in SEISMIC_EVENT
- **Business Rule:** All events have a location (NOT NULL constraint)

Relationship 4: SEISMIC_EVENT → MAGNITUDE_MEASUREMENT

- **Cardinality:** One-to-Many (1:M)
- **Description:** Each event can have one or more magnitude measurements. Each measurement belongs to exactly one event.
- **Implementation:** event_id foreign key in MAGNITUDE_MEASUREMENT
- **Business Rule:** All events have at least one magnitude measurement

Relationship 5: MAGNITUDE_MEASUREMENT → MAGNITUDE_TYPE

- **Cardinality:** Many-to-One (M:1)
- **Description:** Each measurement uses exactly one magnitude scale. Each scale is used in many measurements.
- **Implementation:** magnitude_type_id foreign key in MAGNITUDE_MEASUREMENT
- **Business Rule:** All measurements must specify scale type (NOT NULL constraint)

Relationship 6: SEISMIC_EVENT → QUALITY_METRICS

- **Cardinality:** One-to-One (1:1)
- **Description:** Each event has exactly one set of quality metrics. Each metric set belongs to exactly one event.

- **Implementation:** event_id foreign key in QUALITY_METRICS (with UNIQUE constraint)
- **Business Rule:** Quality metrics are optional (many historical events lack this data)

Normalization Analysis

Our database design achieves Third Normal Form (3NF) by organizing data so that each piece of information is stored in one logical place without redundancy.

How we achieved 3NF:

We extracted repeating categorical values into separate lookup tables, which eliminates redundancy and ensures each attribute depends only on its table's primary key.

EVENT_TYPE Lookup Table

- Instead of storing "earthquake" in 100,000+ rows, we store it once and reference it with an ID
- Saves space and ensures consistency across the database

MAGNITUDE_TYPE Lookup Table

- Instead of repeating magnitude codes like "mb" and "mw" thousands of times, we store each code once
- Allows us to add descriptions and formulas for each magnitude scale

REVIEW_STATUS Lookup Table

- Instead of storing status values like "reviewed" in every event, we store each status once
- Makes it easy to add new status types in the future

QUALITY_METRICS Separation

- Quality measurements are separated into their own table even though they relate 1:1 with events
- Keeps the main SEISMIC_EVENT table focused and handles missing data better (older events often lack quality metrics)

No denormalization was needed - the design performs well in its normalized state.